

Assignment 3:

Question 1: Set of N bytes stored from E501H. The value of N is stored in E500H. Write a program to store these data bytes from m/m location ~~from~~ E600H if D₀ or D₇ is 1.

Solution:N $\Rightarrow$ E500H

Result :- from E600H

Array $\Rightarrow$ from E501HAlgorithm:HL $\leftarrow$ E500H~~from~~ C $\leftarrow$ [HL] (C $\leftarrow$ N)~~DE~~ DE $\leftarrow$ E600H (Result address)HL $\leftarrow$ HL + 1 (HL now points to first number)Loop: A $\leftarrow$ [HL]if (A & 81H $\neq$ 0)

goto SKIP

A $\leftarrow$ [HL]

- store A to address pointed by D.

DE $\leftarrow$ DE + 1SKIP: HL $\leftarrow$ HL + 1C $\leftarrow$ C - 1if (C $\neq$ 0)

goto loop

HLT

Code:

<u>opcode</u>	<u>hex code</u>	<u>Description</u>
LXI H, E500H	21 00 E5	Load immediate HL
MOV C, M	4E	
LXI D, E600H	11 00 E6	
INX H	23	
LOOP: MOV A, M	7E	
ANI 81H	E6 81	AND immediate 81H with 81H A
CPI 00H	FE 00	Compares A with 00H
JC SKIP	CA 16 E0	} if $A \leq 00H$ go to skip.
JZ SKIP	DA 16 E0	
MOV A, M	7E	
STAX D	12	Store accumulator to D.
INX D	13	increment DE
SKIP: INX H	23	increment HL
DCR C	0D	Decrement C
JNZ LOOP	C2 08 E0	
HLT	HLT	

Result: N = 5

Array = A1, 66, E0, 14, B2

Result:A1 = 1010 0001E0 = 1110 0000B2 = 1011 0010

Question - 2:

There are N bytes stored from m/m location $E501H$. The value of N is stored in $E500H$. Write a program to find the sum of integers whose LSB and MSB are 1. Store result in $E600H$ and $E601H$.

Solution:Algorithm:

$HL \leftarrow E500H$ (points to data)

~~HL~~ $D \leftarrow [HL]$ ($0 \leftarrow N$)

$B \leftarrow 00H$ $C \leftarrow 00H$

$HL \leftarrow HL + 1$ (points to array)

LOOP: $A \leftarrow M[HL]$ (points to element of array)

if ($A \& 81H \neq 81H$)

goto SKIP

$A \leftarrow B$ (B has the sum)

$A = A + M[HL]$ (Add current element to A)

if (no carry)

goto SUM

$C \leftarrow C + 1$ (Handles carry)

SUM: $B \leftarrow A$ (Sum stored in B)

SKIP: ~~HL~~ $HL \leftarrow HL + 1$ (HL points to next element)

$D \leftarrow D - 1$

if ($D \neq 0$)

goto LOOP

Store B to $E600H$

Store C to $E601H$

HLT.

Code:

<u>opcode</u>	<u>hexcode</u>	<u>Description</u>
LXI H, E500H	21 00 E5	
MOV D, M	56	Store value of M
MVI B, 001H	06 00	
MVI C, 001H	0E 00	
INX H	23	
LOOP: MOV A, M	7E	Element of array to accumulator
ANI 81H	E6 81	Bitwise and to mask LSB, MSB
CPI 81H	FE 81	Compare with 81H
JNZ SKIP	C2 18 E0	Skips addition if condition not met.
MOV A, B	78	
ADD M	86	Adds the numbers
JNC SUM	02 17 E0	} carry handling
INRC	06	
SUM: MOV B, A	47	Storing sum
SKIP: INX H	23	
DCR D	15	
JNZ LOOP	C2 09 E0	
MOV A, B	78	} Storing data to m/m
STA E600H	32 00 E6	
MOV A, C	79	
STA E601H	32 01 E6	
HLT	76	

Result: N = 5 data = A1, 66, 81, 12, B2

Output: (122)_H = (A1)_H + (81)_H

Question 3:

Write an 8085 program to generate N^{th} fibonacci numbers using functions and store it in F050H. The value of N is F060H.

Algorithm:

$A \leftarrow N$

$B \leftarrow 01H$

$D \leftarrow 00H$

$C \leftarrow N$

$C \leftarrow C - 1$

if ($C == 0$)

goto SKIP

CALL Subroutine SUB

SKIP: ~~MO~~ $A \leftarrow B$

Store A to F050H

HLT

Subroutine SUB:

SUB: NOP

LOOP: ~~MO~~ $A \leftarrow D$

ADD B to A

$D \leftarrow B$

$B \leftarrow A$

$C \leftarrow C - 1$

if ($C \neq 0$)

goto LOOP;

RETURN

Code:

<u>opcode</u>	<u>hexcode</u>	<u>Description</u>
LDA E060H	3A 60 E0	
MVI B, 01H	06 01	
MVI D, 00H	16 00	
MOV C, A	4F	
DCR C	0D	
JZ SKIP	CA 0F E0	meets condition when N=1
CALL SUB	CD 14 E0	calls subroutine SUB
SKIP: MOV A, B	78	B = 2 ^{Nth} element of fibonacci series
STA F050H	32 50 E0	
HLT	76	

SUB: NOP

LOOP: MOV A, D	7A	} calculates n th fibo number.
ADD B	80	
MOV D, B	50	
MOV B, A	47	
DCR C	0D	
JNZ LOOP	C2 14 E0	
RET	C9	Returns to calling address.

Result:

$$N = 7$$

$$\text{Output: } F(7) = 13 = (0D)_{16}$$

Question-4

Write a program to transfer a block of bytes of size N from locations 1 to location 2, when the size of overlap between the two locations is defined by M . The values of N and M are stored in $E200H$ and $E201H$ respectively.

$N \Rightarrow E200H$

data from $\Rightarrow E100H$.

$M \Rightarrow E201H$

Algorithm:

$HL \leftarrow E200H$ (points to value of N)

$D \leftarrow N$ $A \leftarrow N$

$HL \leftarrow HL + 1$ (points to value of M)

SUBTRACT $[HL]$ from A ($A = N - M$)

$HL \leftarrow E100H$ (points to first element of array)

$C \leftarrow A$ ($C = N - M$)

$B \leftarrow 00H$ ($BC = 0(N-M)H$)

$DAD B$ ($HL = HL + (N-M)$)

(HL now points to start of new array)

$C \leftarrow D$ (~~0~~ $C = N$)

$DE \leftarrow E100H$ (points to first element of Array)

$DAD B$ ($HL + N$)

$DCX H$ (HL points to end of new array)

$DE \leftrightarrow HL$

$DAD B$

$DCX H$

} DE points to end of new array
 HL points to end of old array.

~~Repeat~~ Move each element

from HL to DE and decrement both.

END.

Teacher's Signature _____

<u>Code</u>	<u>opcode</u>	<u>Hexcode</u>	<u>Description</u>
LXI H, E200H		21 00 E2	
MOV D, M		56	Store value of N
MOV A, M		7E	
INX H		23	
SUB M		96	Evaluate N-M
LXI H, E100H		21 00 E1	
MOV C, A		4F	
MVI B, 00H		06 00	
DAD B		09	Add (N-M) to HL
MOV C, D		4A	move value of N to C
LXI D, E100H		11 00 E1	
DAD B		09	Move HL to end of new ^{array} loop
DCX H		2B	
XCCH		EB	Exchange HL and DE
DAD B		09	Move HL to end of array.
DCX H		2B	
LOOP: MOV A, M		7E	Load values from initial
STAX D		12	array and store in new
DCX D		1B	array.
DCX H		2B	
DCR C		0D	
JNZ LOOP		C2 17 F0	
HLT		76	

Result: N = 4 E100 onwards: 05, 06, 00, 12
M = 2 → E102 onwards: 05, 06, 00, 12
(N-M=2) output

Question - 5

Flash "BCSE II" in address and data fields, flashing rate of 0.5s.

Solution:

Data: .

Algorithm:

```
SP ← 30FF      (load 30FF to stack pointer)
Call CLEAR Subroutine      (CLEARs display)
                        (using XOR) (A XOR A)
LOOP: A ← 0
      B ← A
      HL ← 2050H      (Point to data)
      Call OUTPUT Subroutine      (Display output)
      A ← 01H
      B ← 00H
      HL ← 2054H      (Points to second part of data)
      CALL OUTPUT Subroutine
      DE ← 0000H
      CALL DELAY Subroutine      (Delays execution depending on
                                value of DE)

      Call CLEAR Subroutine
      DE ← 0000H
      JMP LOOP
END
```

Code:

<u>Opcode</u>	<u>Hex code</u>	<u>Description</u>
LXI SP, 30FFH	31 FF 30	
CALL CLEAR	CD 47 03	CLEAR subroutine called
LOOP: XRA A	AF	Sets A to 0
MOV B, A	4F	
LXI H, 2050H	21 50 20	} display first four values in data field.
CALL OUTPUT	CD 00 05	
MVI A, 01H	3E 01	
MVI B, 00H	06 00	
LXI H, 2054H	21 54 20	} for displaying last two values in data field.
CALL OUTPUT	CD 00 05	
LXI D, 0000H	11 00 00	} Introducing delay after displaying.
CALL DELAY	CD BC 03	
CALL CLEAR	CD 47 03	} delay after clearing.
LXI D, 0000H	11 00 00	
CALL DELAY	CD BC 03	
JMP LOOP	11 00 00	
HLT	76	

Note: The data to be flashed on the screen is determined by the hex values stored in data memory.